Introduction to Multiplication: Equal Groups & Arrays

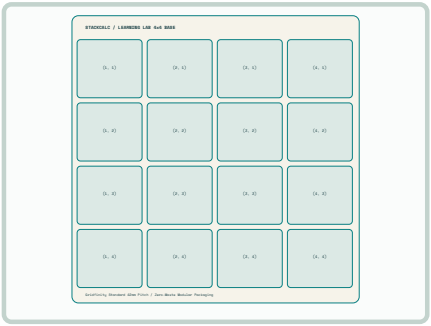
Tactile Concrete-Pictorial-Abstract (CPA) Station Guide • CCSS.MATH.3.OA.A.1

Student Name: _____ Date: _____ **Model: SC-6** **Check Score: ___ / 4**

STUDENT SELF-CHECK RULE (NO CHEATING):
Build with blocks and solve on paper first! Use your SC-6 calculator ONLY to verify answers on the 4-line LCD screen.

STATION LAB TUTORIAL: BENCH SCHEMATIC & CPA PROGRESSION

1. 2D Bench Blueprint



Kit: Gridfinity Baseplate

2. Desk Setup (Concrete -> Pictorial)

Physical Bench Setup:
Snap cubes into 3 equal towers of 4 cubes each on the Gridfinity dock.

Sketch Model on Paper:
Draw an array with 3 rows and 4 columns.

3. Step-by-Step LCD Stack Trace (SC-6)

MODE:	123	Base-10 Arithmetic	
1.	4	X: 4	Number of groups (4)
2.	ENTER	Y: 4 | X: 4	Stack lift
3.	6	Y: 4 | X: 6	Size per group (6) i
4.	×	X: 24	Product: 4 × 6 = 24

EXPECTED LCD X: 24 cubes

1 6 equal groups of 4 markers. Total markers?

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

2 5 rows of 3 chairs. Total chairs?

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

3 8 packages with 2 books each. Total books?

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

4 9 teams with 5 players each. Total players?

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

Station Diagnostic & Solution Key: Unit 2

Introduction to Multiplication: Equal Groups & Arrays • Manipulative Setup, Diagnostic Rubric & Verification Flow

MANIPULATIVE HARDWARE STATION INVENTORY & 3D PRINT PREPARATION

Station Kit: Gridfinity 4x4 Coordinate Baseplate & Cubes

CAD Blueprint: gridfinity_baseplate

Target Device: SC-6 Elementary

CAD / STL Source File:

Education/scad/lib/gridfinity.scad

Standard: Gridfinity 42mm Pitch / 0.6mm Inset Markings

Recommended Filaments:

Charcoal 4x4 Baseplate & Teal Cubes (PLA)

#	Math Problem Prompt	Expected Math Answer	Self-Check Keystrokes & RPN Stack
1	**6 equal groups of 4 markers. Total markers?** Standard: CCSS.MATH.3.OA.A.1	24	[6] [ENTER] [4] [×]
2	**5 rows of 3 chairs. Total chairs?** Standard: CCSS.MATH.3.OA.A.1	15	[5] [ENTER] [3] [×]
3	**8 packages with 2 books each. Total books?** Standard: CCSS.MATH.3.OA.A.1	16	[8] [ENTER] [2] [×]
4	**9 teams with 5 players each. Total players?** Standard: CCSS.MATH.3.OA.A.1	45	[9] [ENTER] [5] [×]

TEACHER DIAGNOSTIC GUIDE: COMMON MISCONCEPTIONS & REMEDIATION

Key Misconception to Target During Station Observation:

Clarify that 3 groups of 4 and 4 groups of 3 have the same product but different physical orientation.

How the Manipulatives & Calculator Foster Self-Correction:

Students solve with pencil and physical blocks first. When they self-check on the SC-6, any discrepancy between their written answer and the calculator's 4-line LCD immediately prompts them to re-inspect their physical bench model or regrouping, turning errors into productive struggle without teacher intervention.

Station Facilitation & Pacing Tip:

Keep 3D printed trays docked on standard Gridfinity baseplates to prevent loose parts on student desks. Encourage students to articulate their CPA steps: name the block representation (Concrete), point to their drawn tape/bond model (Pictorial), and verify register values (Abstract).

CLASSROOM SCORING & STATION MASTERY RUBRIC

Exemplary (18–20 pts):	Complete, accurate manipulative sketches; exact math answers; independently verified self-check on LCD stack.
Proficient (14–17 pts):	Correct math answers; minor sketch omissions; verified on calculator with minimal teacher prompting.
Developing (10–13 pts):	Understands problem concept but made computational slips; successfully self-corrected using calculator LCD.
Intervention (<10 pts):	Relied on calculator before solving on paper, or struggled with underlying physical manipulative models.